



Antihypertensive Agents

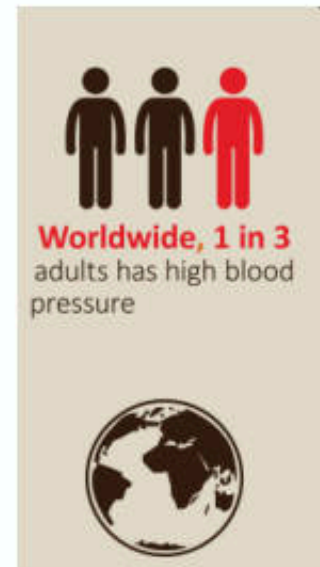
Krisada Sakchaisri

วัตถุประสงค์เชิงพฤติกรรม

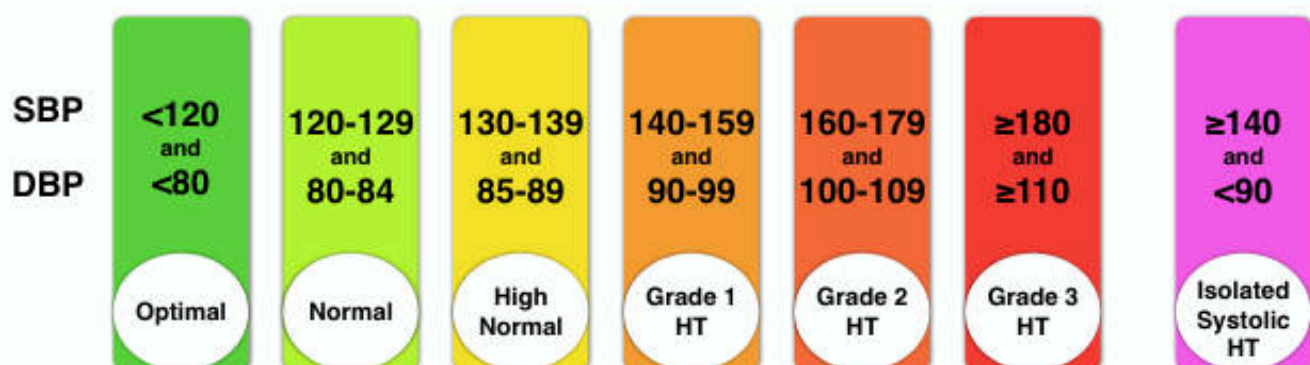
1. สามารถจำแนกและอธิบายกลไกการออกฤทธิ์ของยาลดความดันโลหิตในกลุ่มต่างๆได้
2. สามารถบรรยายอาการไม่พึงประสงค์ และข้อควรระวังอันเกี่ยวเนื่องจากฤทธิ์ทางเภสัชวิทยาของยาของยาลดความดันโลหิตสูงได้

Hypertension

- ♦ The most common cardiovascular disease
(Major public health problem)
- ♦ Increasing with age
(50% of people between the ages of 60-69 years old have hypertension)
- ♦ Simply defined as a sustained increase in blood pressure $\geq 140/90$ mmHg

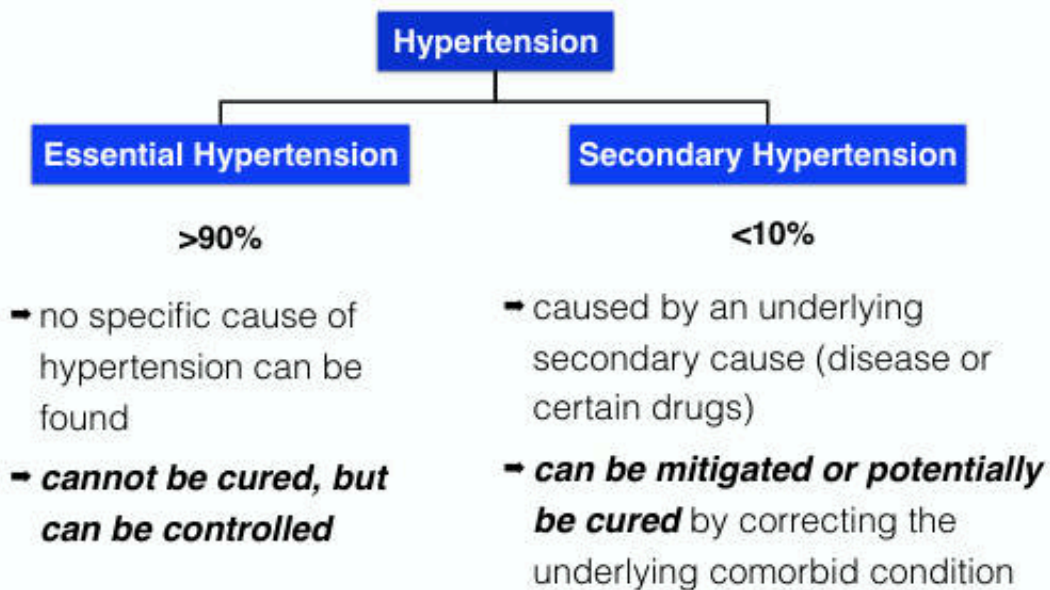


Thai Guideline (2019) & ESH/ESC Guideline 2018

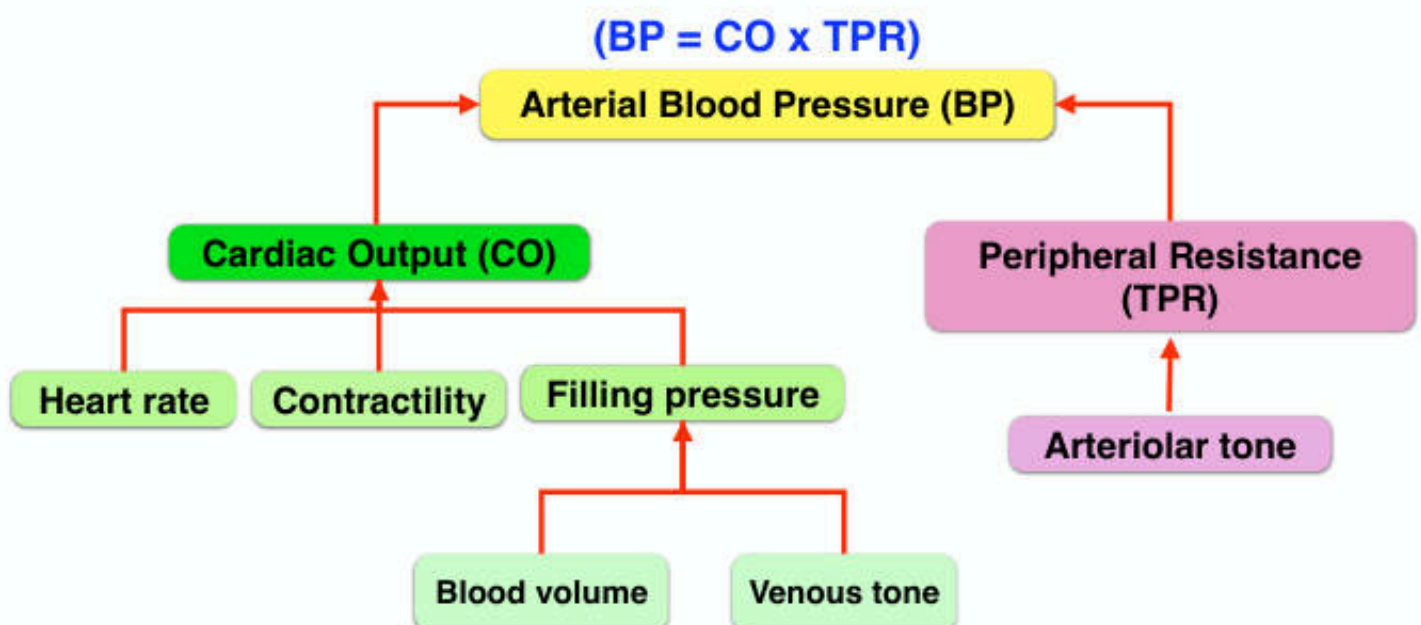


Based on **repeated, reproducible** measurements of elevated blood pressure

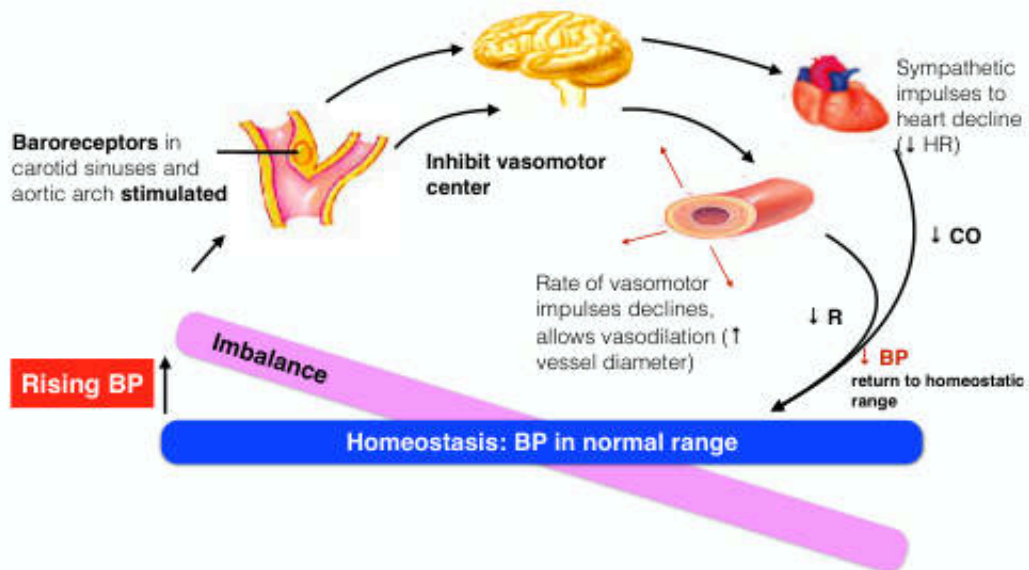
Etiology



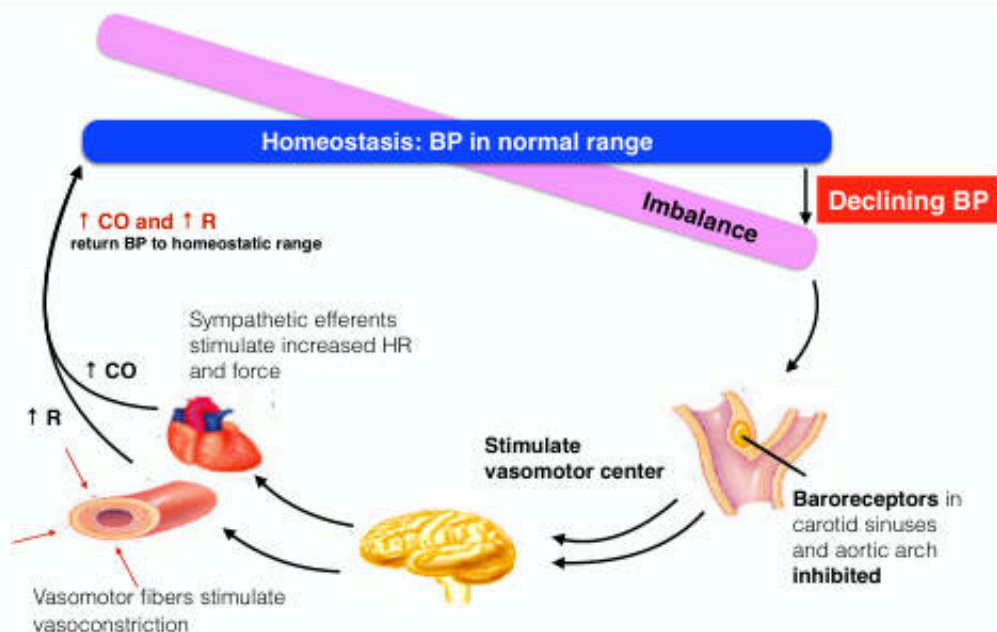
Major Factors Influencing Blood Pressure



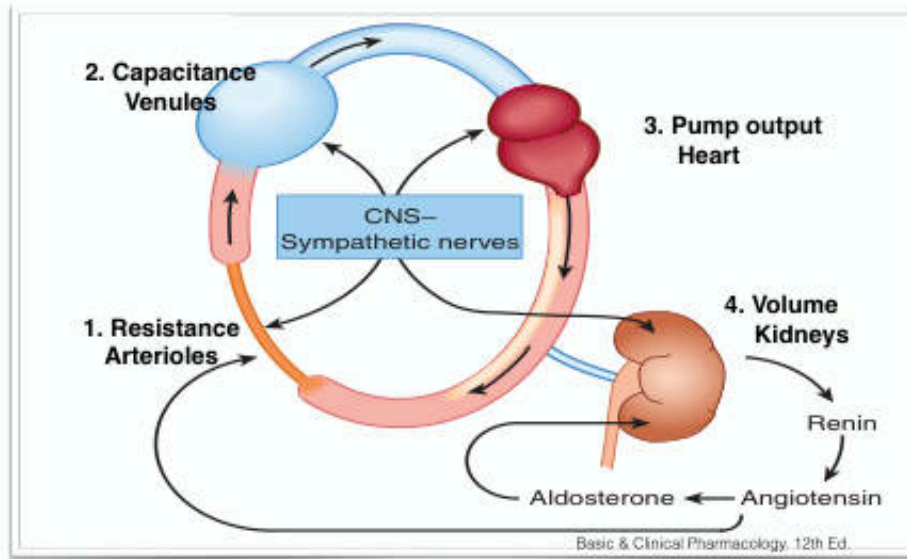
Regulation of Blood Pressure by Baroreceptor Reflex



Regulation of Blood Pressure by Baroreceptor Reflex



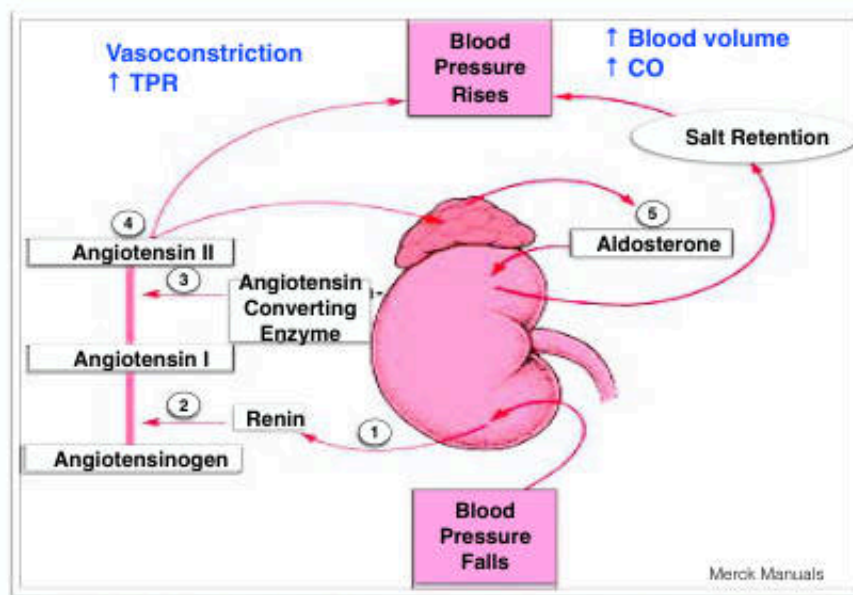
Normal Regulation of Blood Pressure



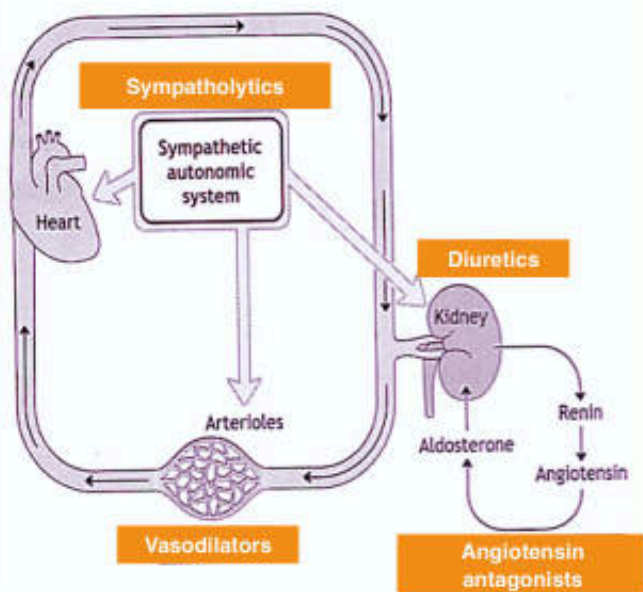
Blood pressure is maintained by moment-to-moment regulation of cardiac output (CO) and peripheral vascular resistance (PVR)

Regulation of Blood Pressure by Kidney

kidney is primarily responsible for long-term blood pressure control



Classification of Antihypertensive Drugs



1. Diuretics

: ↓ Na⁺, ↓ blood volume

2. Sympatholytic agents

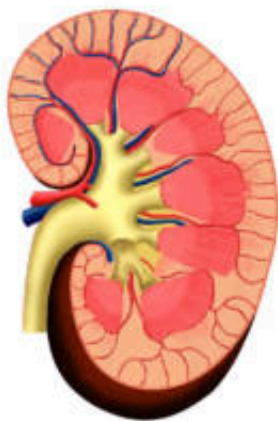
: ↓ TPR (resistance), ↓ cardiac function,
↑ Capacitance (venous pooling)

3. Direct vasodilators

: Relaxing vascular smooth muscle
↓ TPR, ↑ capacitance

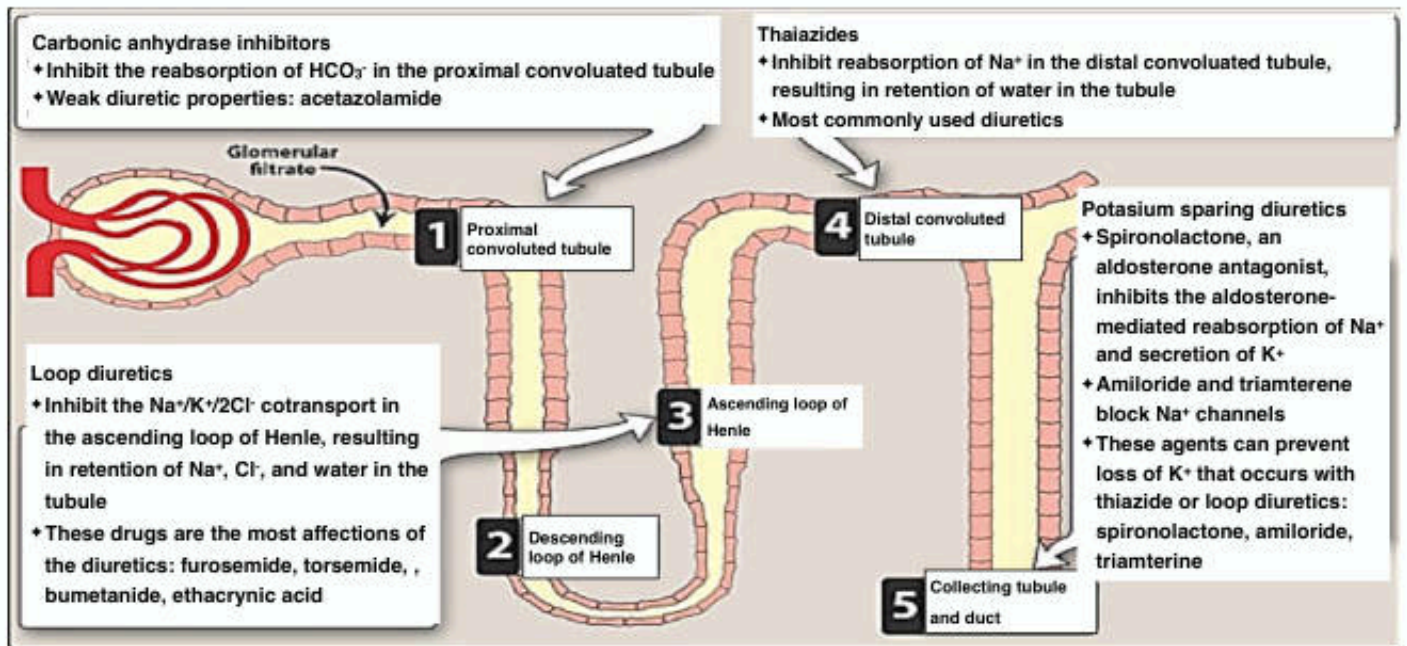
4. Agents that block production or action of angiotensin

: ↓ TPR, ↓ blood volume (potentially)



Diuretics

Sites of Action of Diuretic Drugs



Lippincott's Illustrated Reviews Pharmacology

Diuretics



Inhibition of sodium resorption



Thiazides

- hydrochlorothiazide (HCTZ)
- chlorothiazide
- chlorthalidone
- indapamide

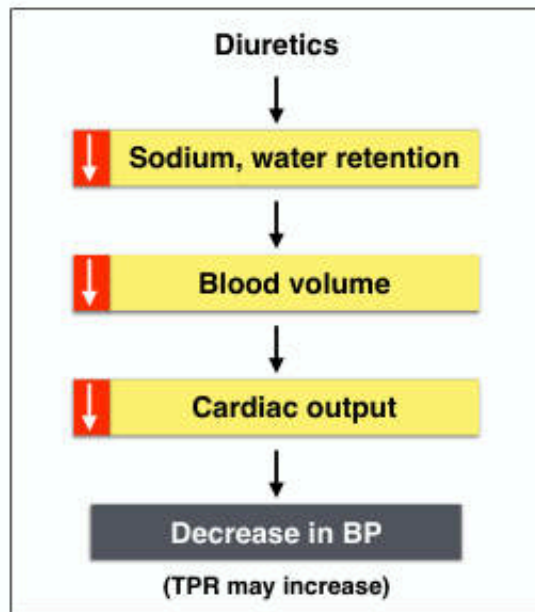
Loop diuretics

- furosemide
- bumetanide
- torsemide
- ethacrynic acid

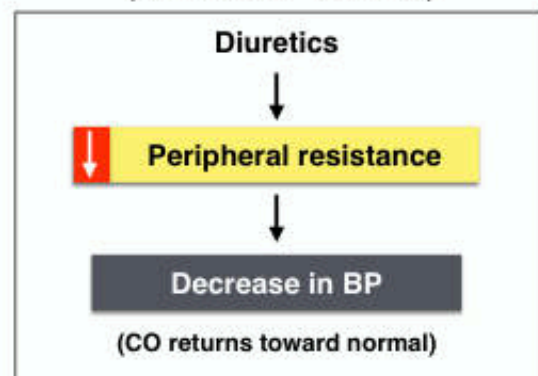
K⁺-sparing diuretics

- amiloride
- triamterene
- spironolactone
- eplerenone

Initial Effects



Long-Term Effects (after 6-8 weeks)

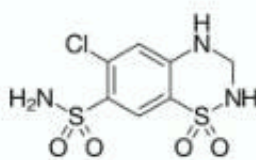


Sodium is believed to contribute to vascular resistance by

- ✦ increasing vessel stiffness
- ✦ neural reactivity

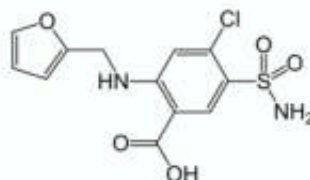
Uses of Diuretics

↓ 10-15 mmHg



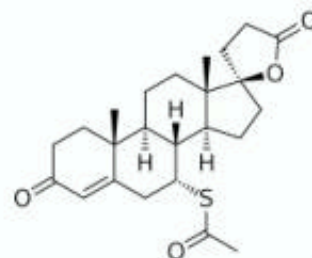
HCTZ

★ mild or moderate HT



Furosemide

★ severe HT



Spirolactone

★ useful to avoid excessive potassium depletion

Adverse Effects of Diuretics

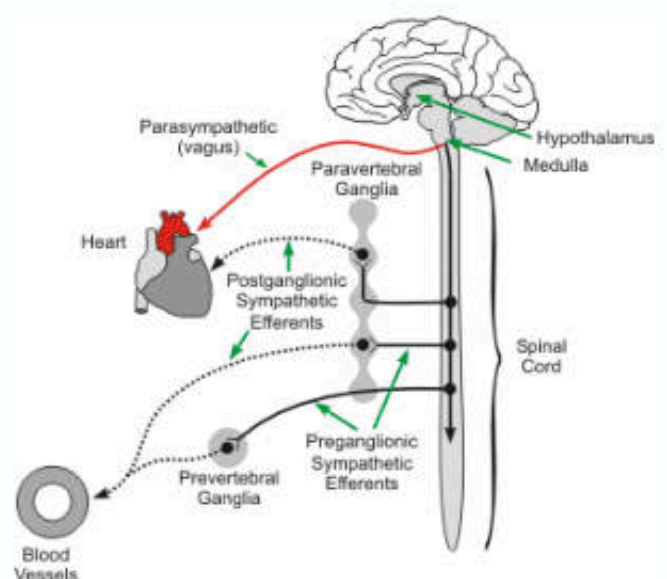
Electrolyte imbalance



- ✦ **hypokalemia** (except for K⁺-sparing diuretics)
- ✦ **hyperkalemia** (K⁺-sparing diuretics), particularly in patients taking **ACE inhibitors** or angiotensin receptor blockers (**ARBs**)

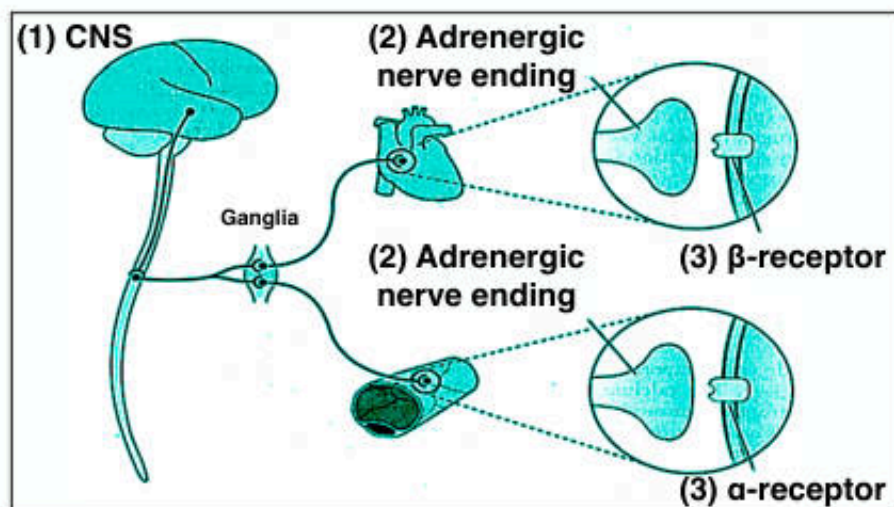
- ✦ May also cause magnesium depletion
- ✦ May impair glucose tolerance, and increase serum lipid concentrations
- ✦ Diuretics increase uric acid concentrations and may precipitate gout

Sympatholytic Agents

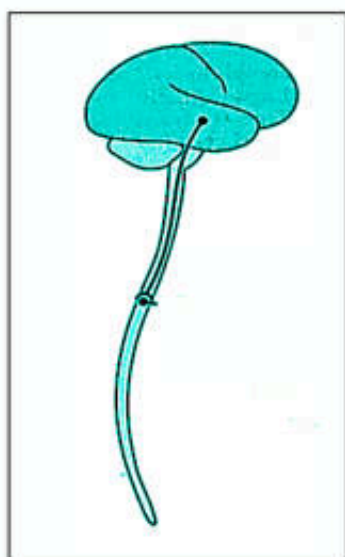


Sympatholytic Agents

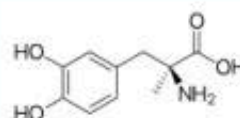
1. Centrally acting adrenergic agents
2. Adrenergic neuron blocking agents
3. Adrenoceptor blockers



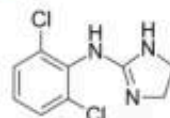
Centrally Acting Sympatholytic Agents



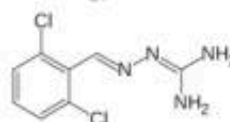
★ **Methyldopa**



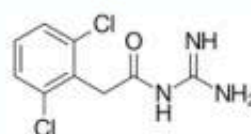
★ **Clonidine**



★ **Guanabenz**



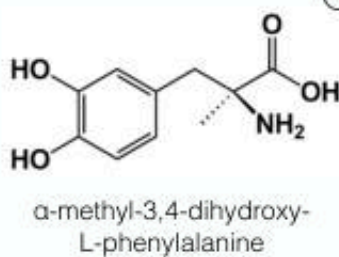
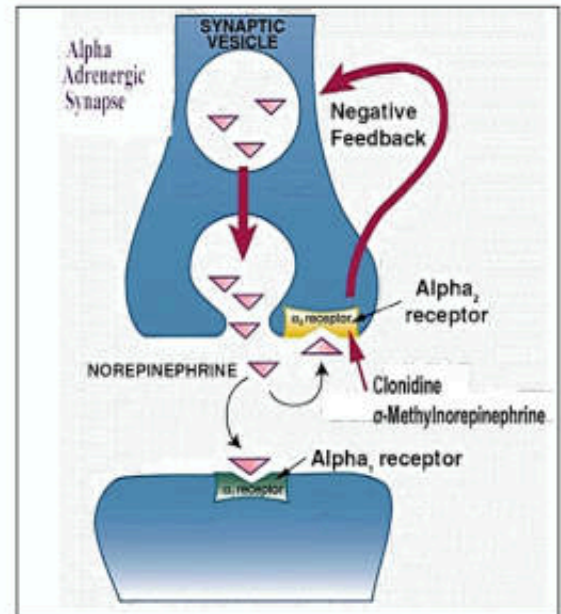
★ **Guanfacine**



Centrally Acting Adrenergic Neuron Blockers

- ★ Act in the brain by binding to **presynaptic α_2 -adrenergic receptor**
- ★ **Reduce sympathetic outflow** from **vasomotor center** in the brain stem

↓ Cardiac output, ↓ TPR



Methyldopa (an analog of L-dopa)

↓ L-aromatic amino acid decarboxylase

↓ Dopamine β -oxidase

α -Methylnorepinephrine

Substitute NE at adrenergic nerve vesicle

Acting as an agonist at **presynaptic α_2 -adrenergic receptors**

↓ NE release, thereby ↓ vasoconstrictor adrenergic signals

↓ TPR

with a variable reduction in HR and CO

Methyldopa

↓ ~20/13 mmHg
mild HT

↓ ~38/22 mmHg
severe HT



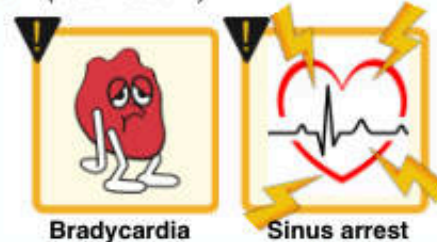
- ❖ Widely used in the past but is now used primarily for **hypertension during pregnancy**, *based on its effectiveness and **safety***
- ❖ Generally has been used in association with diuretics due to an **expanded plasma volume**

Methyldopa: Adverse Effects

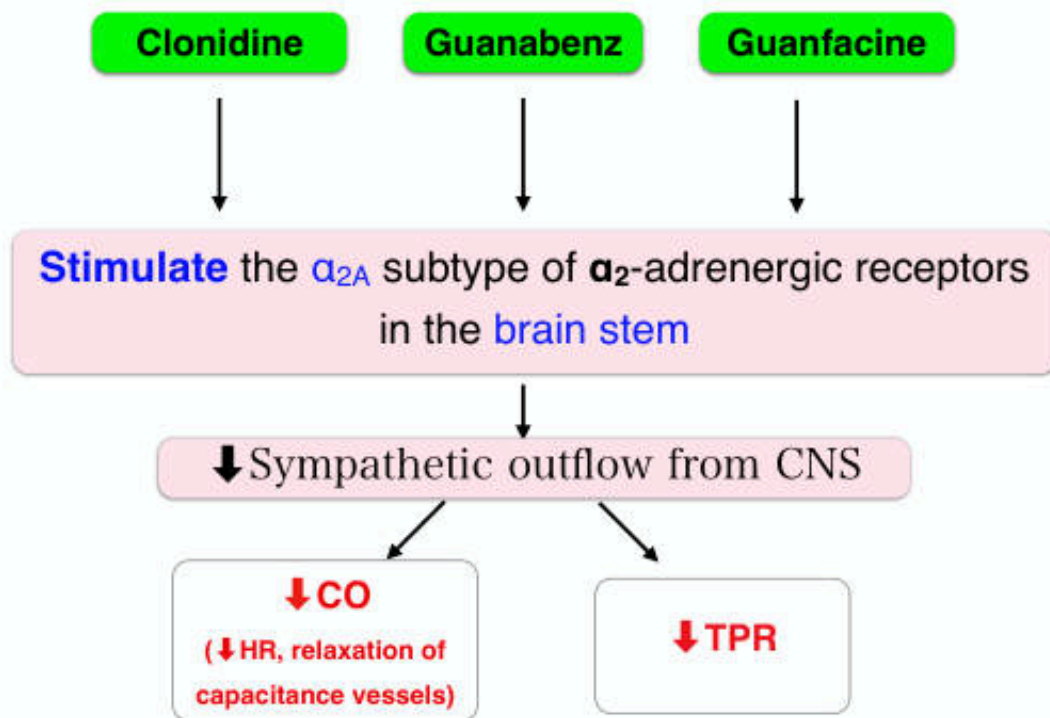


- ❖ The most common undesirable effect of methyldopa is **sedation**, *particularly at the onset of treatment...(transient)*

- ❖ May precipitate bradycardia and sinus arrest



- ❖ Other adverse effects: *dryness of mouth, parkinsonian sign, hyperprolactinemia*
- ❖ At least 20% of patients, who receive methyldopa for a year, develop a (+) Coombs test (red blood cell antibody screening).....*reversible*



Clonidine

↓ **~25/15 mmHg** (maximally effective and tolerated dose)

- ❖ **Not a leading option** for monotherapy of hypertension
- ❖ Effectively lower BP in some patients who have not responded adequately to combination therapy of other agents
- ❖ **Limitation:** the paucity of information about their efficacy in reducing the risk of cardiovascular consequences of hypertension

Clonidine: Adverse Effects



✦ The common adverse effects are **dry mouth** and **sedation**

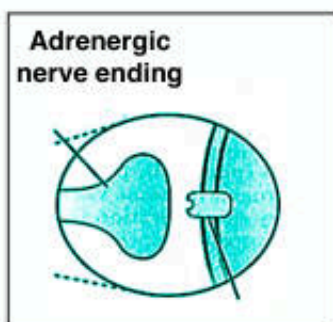


✦ Other adverse effects: *GI disturbances, weight gain, nightmares, insomnia, anxiety, depression, urinary retention, and impotence*



✦ **Suddenly discontinuation** of these drugs can result in life-threatening **hypertensive crisis** mediated by increased sympathetic nervous activity

Adrenergic Neuron Blocking Sympatholytic Agents

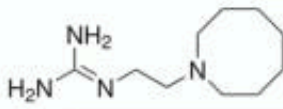


preventing normal physiologic
release of NE from
postganglionic sympathetic
neurons

★ Guanethidine

★ Reserpine

Guanethidine



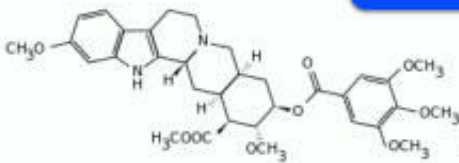
Replace NE at transmitter vesicles
"False Neurotransmitter"

Gradual depletion of NE stores
at nerve ending

Reduction of both arteriolar
and venous tone
(↓TPR)

- ✦ Neuronal uptake, *NE transporter (NET)*, is necessary for the hypotensive activity
- ✦ Drugs that block the catecholamine uptake process attenuate this effect
: *cocaine, amphetamine, and tricyclic antidepressants (TCAs)*
- ✦ Adverse effects:
postural hypotension
diarrhea
sexual dysfunction (impaired ejaculation)

Reserpine



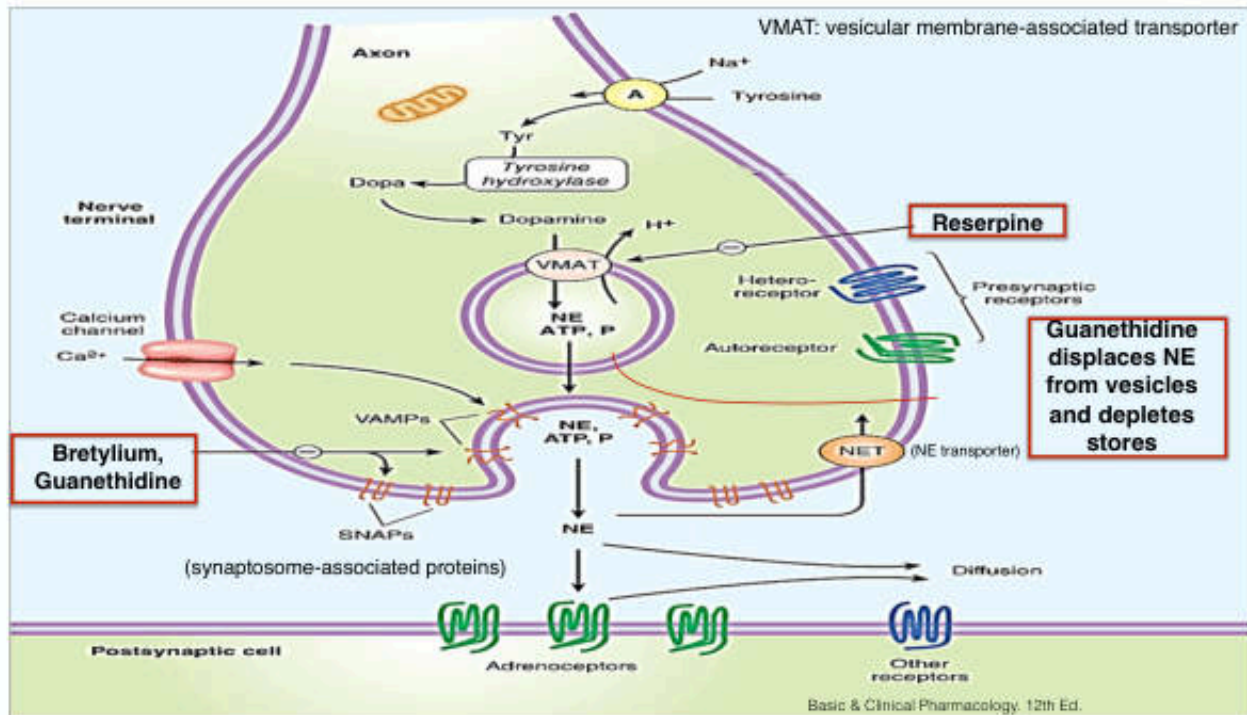
Block the uptake and store of
biogenic amines into synaptic
vesicles

Depletion of NE, DA, and 5-HT
in both central and peripheral
neurons

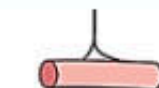
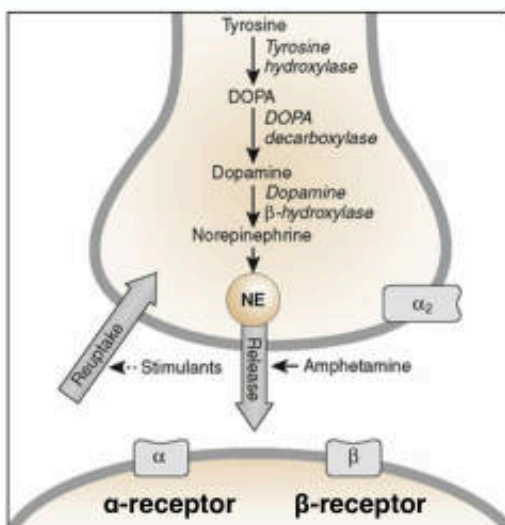
↓CO and ↓TPR

- ✦ An alkaloid extracted from the roots of *Rauwolfia serpentine*
- ✦ The effects are probably by interfering with the vesicular membrane-associated transporter (*VMAT*)
- ✦ Adverse effects:
most unwanted effects result from actions on
: *the brain(sedation, lassitude, depression)*
: *GI tract (mild diarrhea, GI cramps, and increased gastric acid secretion)*

Sites of Action of Guanethidine and Reserpine

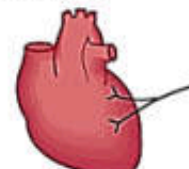


Adrenoceptor Blockers



α-receptor of vessels

β-receptor of heart



β-receptor of juxtaglomerular cells that release renin

Adrenoceptor Blockers

α -Adrenoceptor antagonists

- ✦ Prazosin, doxazosin, terazosin
- ✦ Block $\alpha_1 \gg \gg \gg \alpha_2$
- ✦ $\downarrow$ BP, less reflex tachycardia

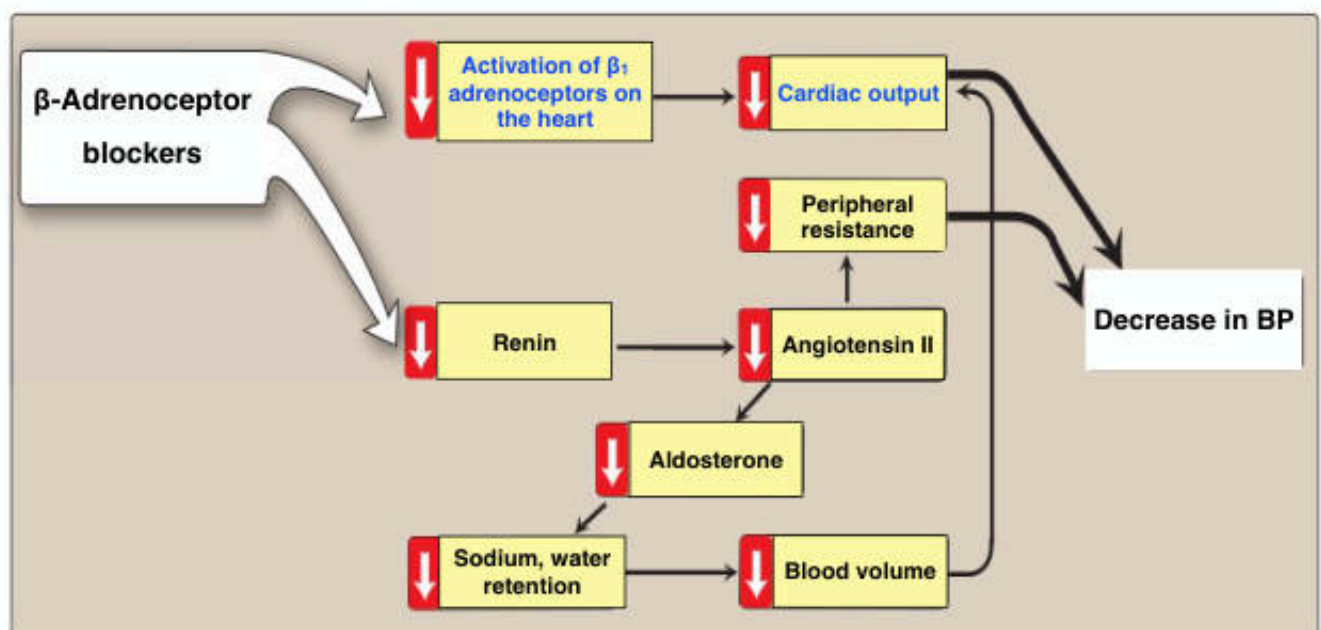
Mixed $\alpha\beta$ -Adrenoceptor antagonists

- ✦ Labetalol, carvedilol
- ✦ Block $\beta_1 = \beta_2 \geq \alpha_1 > \alpha_2$
- ✦ $\downarrow$ BP with limited HR increased

β -Adrenoceptor antagonists

- ✦ **Non-selective (block β_1 and β_2)**
 - Propranolol, nadolol, timolol
 - $\downarrow$ HR, $\downarrow$ BP, and $\downarrow$ renin
- ✦ **β_1 -Selective (block $\beta_1 > \beta_2$)**
 - Metoprolol, atenolol, alprenolol
 - Maybe safer in asthma
 - $\downarrow$ HR, $\downarrow$ BP, and $\downarrow$ renin
- ✦ **β -Blockers with ISA**
 - Pindolol, acebutolol, penbutolol
 - $\downarrow$ BP, moderately $\downarrow$ HR

Action of β -Adrenoceptor Blockers



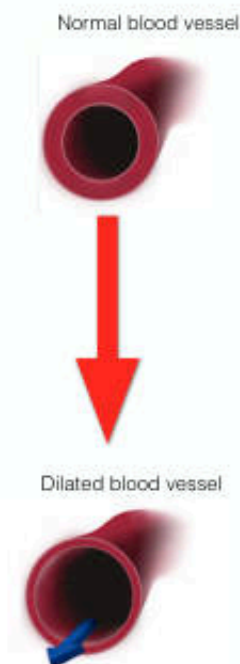
β Adrenergic Blockers

- ❖ Should be **avoid** in patients with:
 - **Asthma**
 - **SA or AV nodal dysfunction**
 - *In combination with other drugs that **inhibit AV conduction***
- ❖ The **risk of hypoglycemic reactions** may be increased in diabetics taking **insulin**
- ❖ **↑ Triglycerides, ↓ HDL** (*ISA group having little or no effect*)
- ❖ Sudden discontinuation can produce "**withdrawal syndrome**"
 - *dosage should be taper gradually over 10-14 days*
- ❖ **NSAIDs** such as indomethacin can **blunt** the antihypertensive effect of propranolol and probably other β receptor antagonists

Vasodilators

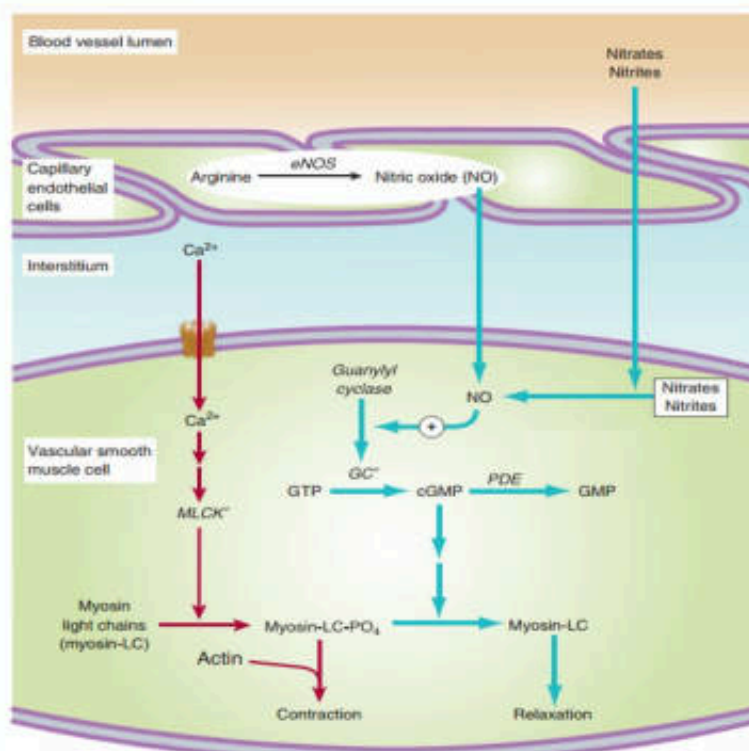


Vasodilators



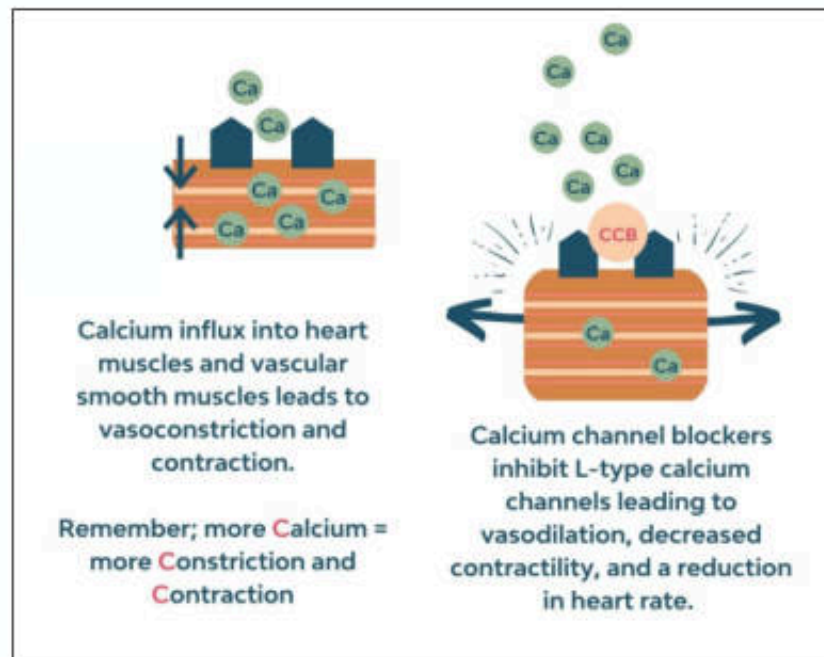
Drugs	Mechanism
Hydralazine	Directly relaxes arteriolar smooth muscle : the precise mechanism of action is not fully understood : may inhibit IP ₃ -induced Ca ₂ ⁺ release from intracellular storage
Nitrates & Nitrites : nitroglycerine : isosorbide dinitrate	Release of nitric oxide (NO)
Calcium channel blockers : amlodipine : verapamil, diltiazem	Reduction of calcium influx
Potassium channel openers : Minoxidil, nicorandil	Hyperpolarization of smooth muscle membrane
Dopamine agonist : Fenoldopam (peripheral arteriolar dilator)	Activation of dopamine D ₁ receptor

Nitrates & Nitrite



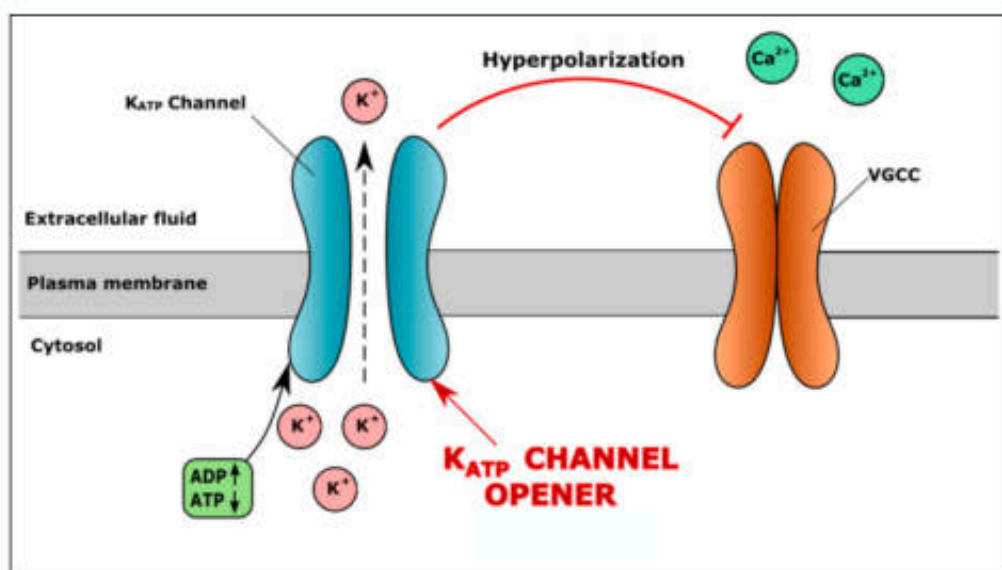
Modified from Basic & Clinical Pharmacology, 12th Ed.

Calcium Channel Blockers



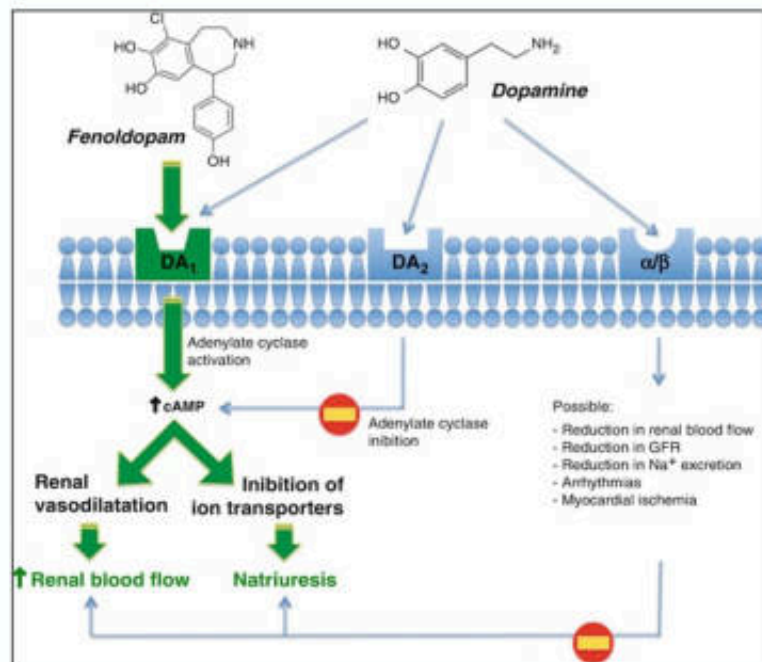
memory-pharm.com

Potassium Channel Openers



Cells 2021, 10(10), 2572.

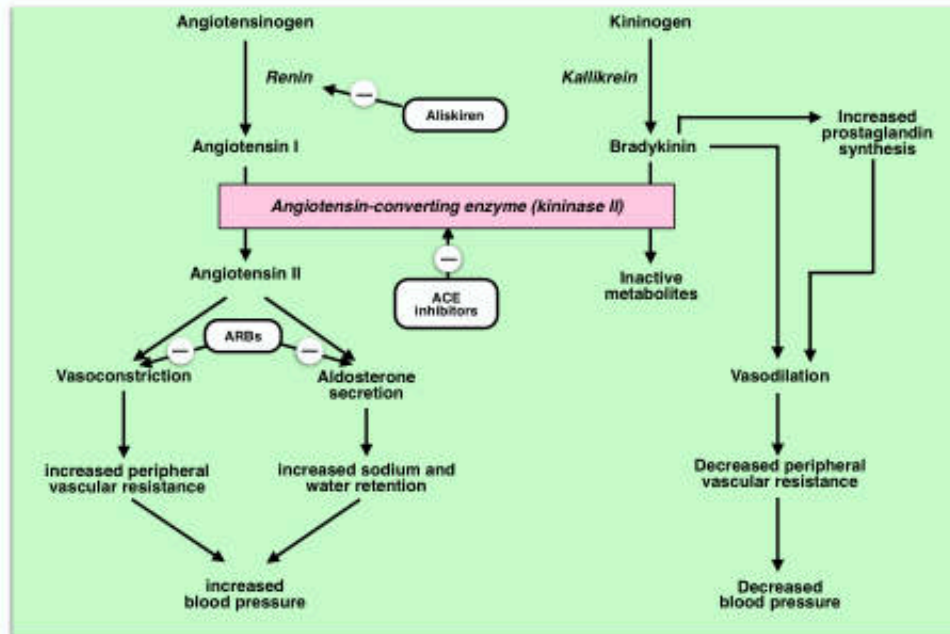
Dopamine Agonist



springernature.com (https://doi.org/10.1007/978-3-319-33429-5_13).

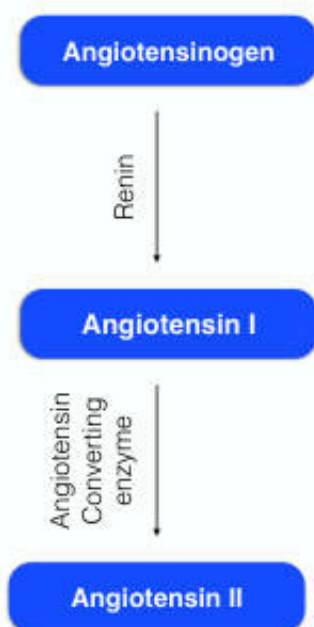
*Agents that Block
Production or Action
of Angiotensin*

Renin-Angiotensin-Aldosterone System



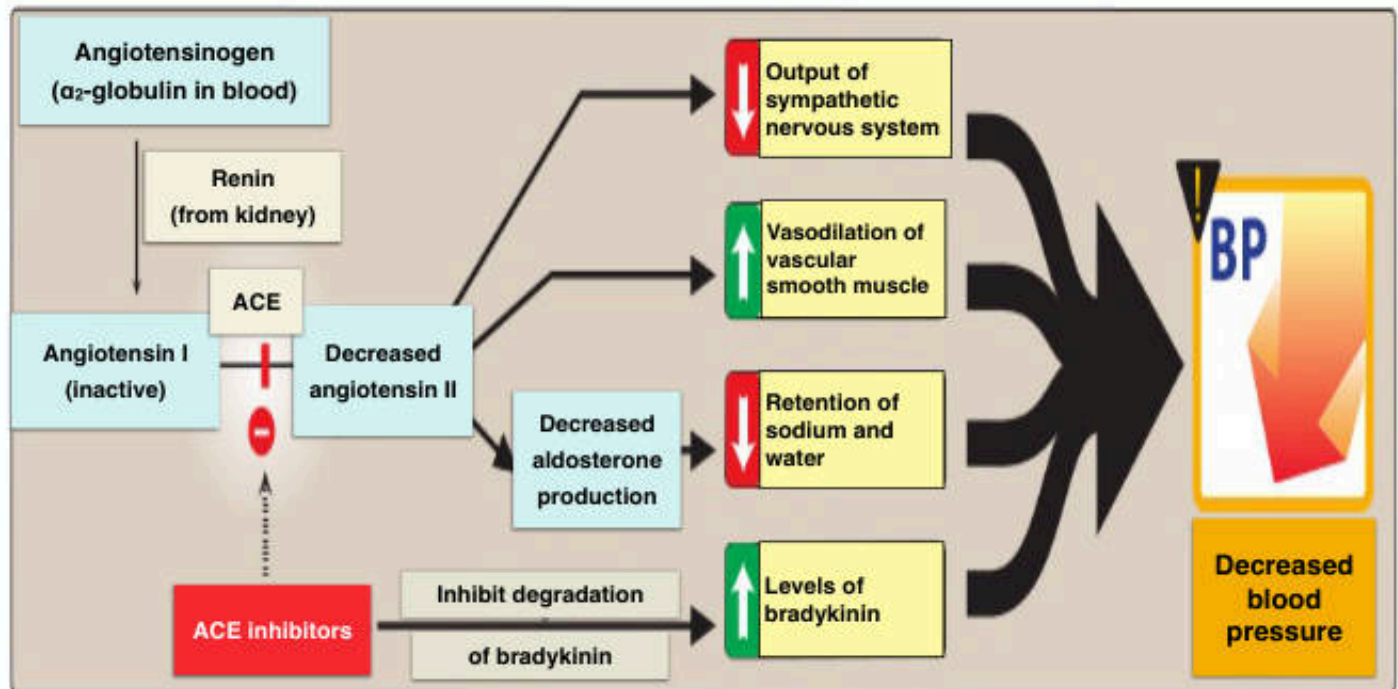
Basic & Clinical Pharmacology, 12th Ed.

Inhibitors of Angiotensin



Subclass	Mechanism
Direct renin inhibitor	
Aliskiren	Inhibit enzyme activity of renin : ↓ angiotensin I and II ↓ aldosterone
Angiotensin-converting enzyme inhibitor (ACEI)	
Captopril Enalapril Lisinopril Ramipril	Reduce angiotensin II levels : ↓ vasoconstriction ↓ aldosterone secretion ↑ increase bradykinin
Angiotensin receptor blockers (ARBs)	
Losartan Candesartan Irbesartan Telmisartan	Block angiotensin II type 1 (AT1) receptor : ↓ vasoconstriction ↓ aldosterone secretion

Action of ACE Inhibitors



ACE Inhibitors & ARBs

- ✦ Severe **hypotension** can occur after **initial doses** in patients who are hypovolemic
- ✦ Other common adverse effects:
 - **Acute renal failure**
 - **Hyperkalemia**
 - **Dry cough** (only with ACE inhibitors)
- ✦ **Contraindicated in pregnancy**
- ✦ Important drug interactions: K^+ supplement or K^+ -sparing diuretics

Vasomotor center (medulla oblongata)

Methyldopa
Clonidine
Guanabenz
Guanfacine



Sympathetic ganglia

Trimethaphan



β-Receptors of heart

β-blockers
(e.g. propranolol)

Sympathetic nerve terminals

Guanethidine
Guanadrel
Reserpine

α-Receptors of vessels

α₁-blockers
(e.g. prazosin)

Vascular smooth muscle

Hydralazine
Minoxidil
Nitroprusside

CCBs

Fenoldopam
Diazoxide

AT1 receptor of vessels

ARBs (e.g. losartan)

Kidney tubules

Thiazides, etc.



β-Receptor of juxtaglomerular cells that release renin
β-blockers (e.g. propranolol)

Angiotensinogen

~~Renin~~

Aliskiren

Angiotensin I

~~ACE~~

ACE inhibitors

Angiotensin II

Consequences of Uncontrolled Hypertension



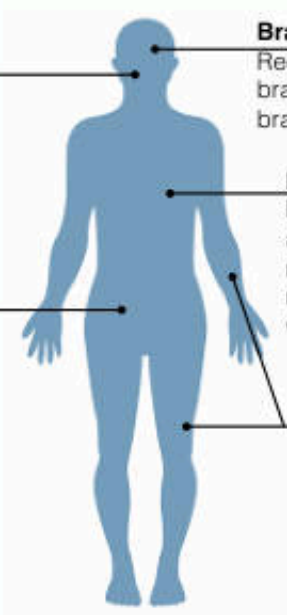
Retinopathy

High BP can damage blood vessels in the retina, resulting in **loss of vision**



Renal failure

Damaged blood vessels in the kidneys can't effectively filter your blood, resulting in a dangerous accumulation of fluid and waste



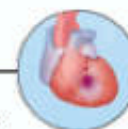
Brain Stroke

Reduced blood supply to the brain can lead to rapid loss of brain function or stroke



Heart attack

Hypertension causes the heart to pump against high BP, making it work harder than necessary. Over time, this causes the heart muscle to thicken, restricting blood flow which can lead to heart failure

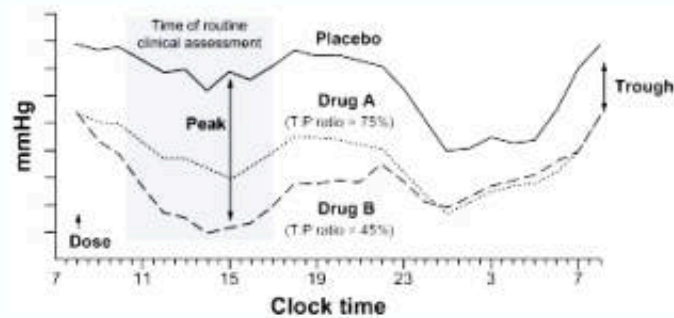


Peripheral vascular disease

Hypertension is a leading cause of atherosclerosis, the artery-narrowing process that can result in heart attack and stroke



Trough:Peak Ratio of Antihypertensive Medications



- ✦ The treatment of hypertension has evolved over time to include medications
 - ➡ Effective at maintaining reductions in average BP > 24 hour period
 - ➡ Being able to do so with medications that can be administered once-a-day
- ✦ FDA proposed that before an antihypertensive medication could be approved for **once-a-day** dosing it would have to have a **trough:peak ratio** value of **greater than 50%**